Robust Self-Tuning Sparse Subspace Clustering

Guangtao Wang*, Jiayu Zhou†, Jingjie Ni‡, Tingjin Luo§, Wei Long¶, Hai Zhen‖, Gao Cong ** and Jieping Ye††
* Comp. Med. and Bioinformatics, University of Michigan, Ann Arbor, MI, USA & Xi'an Jiaotong University, China
†Computer Science and Engineering, Michigan State University, MI, USA
‡Hewlett-Packard Enterprise Company, Boston, MA, USA
§College of Science, National University of Defense Technology, ChangSha, China
¶College of Computer Science, Zhejiang University, Hangzhou, Zhejiang, China
‖ Agency for Science, Technology and Research, Singapore
** School of Computer Science and Engineer, Nanyang Technological University, Singapore
†† Comp. Med. and Bioinformatics, University of Michigan, Ann Arbor, MI, USA

***Abstract*—Sparse subspace clustering (SSC) is an effective approach to cluster high-dimensional data. However, how to adaptively select the number of clusters/eigenvectors for different data sets, especially when the data are corrupted by noise, is a big challenge in SSC and also an open problem in field of data mining. In this paper, considering the fact that the eigenvectors are robust to noise, we develop a self-adaptive search method to select cluster number for SSC by exploiting the cluster-separation information from eigenvectors. Our method solves the problem by identifying the cluster centers over eigenvectors. We first design a new density based metric, called centrality coefficient gap, to measure such separation information, and estimate the cluster centers by maximizing the gap. After getting the cluster centers, it is straightforward to group the remaining points into respective clusters which contain their nearest neighbors with higher density. This leads to a new clustering algorithm in which the final randomly initialized k-means stage in traditional SSC is eliminated. We theoretically verify the correctness of the proposed method on noise-free data. Extensive experiments on synthetic and real-world data corrupted by noise demonstrate the robustness and effectiveness of the proposed method comparing to the well-established competitors.**

***Index Terms*—Cluster Number Estimation; Cluster Center; Sparse Subspace Clustering;**

I. Introduction

In numerous applications of data mining, pattern recognition, image processing and computer vision [1], [2], [3], we are confronted with the problem of clustering high dimensional data points that lie in a union of multiple low dimensional subspaces. Various subspace clustering algorithms have been proposed for such high dimensional data. Most of them find subspaces on the original feature space by specific heuristic search stages [4]. These algorithms usually require that the number of subspaces and the dimensionality of each subspace have to be known in advance. However, estimating these parameters based on the data points is quite difficult [4].

An alternative approach resorts to new representations of the data which embed subspace information automatically. Sparse representation technique provides an effective tool to uncover the low dimensional structures in data [5], [6], [7]. Based on such representations, sparse subspace clustering (SSC) was proposed in [8] and has gained its popularity and success in many applications, such as motion segmentation [9], face clustering [10], [9], etc. It assumes that the data points lie in a union of linear/affine subspaces [8], [10], [9]. Each point can be represented as a linear combination of all other points. By searching for the sparsest combination, it automatically obtains a sparse representation from other points lying in the same subspace. Afterwards, it uses these sparse representations to build an affinity graph of all data points, from which the segmentations of data are explored using spectral clustering.

The standard SSC assumes the data is noise-free. But this assumption is often violated in practice since the real-world data are usually corrupted by noise [9]. The noise results in that the data do not lie perfectly in a union of subspaces [9], [11]. There have been research on extending SSC for handling noisy data [12], in which an approximate sparse representation can be recovered with a bounded error on corrupted data [13]. However, the problem of estimating the correct number of clusters/eigenvectors for SSC is still quite difficult and challenging on the noisy data. As we know, the number of clusters plays a critical role in SSC. It is usually set manually and very limited research has been done on how to automatically set it [9]. The most common method uses a gap in the distribution of the eigenvalues to estimate the cluster number [14], [15], [16]. However, the eigenvalues are sensitive to the noise, and thus, the method is perhaps unreliable to detect the correct cluster number for SSC on corrupted data [9]. SSC employs spectral clustering for data segmentation. There have been a few research on cluster number estimation for spectral clustering [17], [18], [19]. However, the performance of these methods might not be guaranteed for SSC since they are proposed based on the similarity which is computed in original feature space [20], [18], [19], [21], [16], [22], and this is quite different from the similarity used in SSC, which is measured by the sparse coefficients on subspaces.

SSC often applies k-means to divide the data into disjoint segmentations over the eigenvectors [8], [23]. The cluster centers play a key role in k-means. If we can accurately determine the cluster centers, it is straightforward to get the number of clusters, and then can group the rest points into

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correct cluster effectively by assigning them to the group containing their nearest center. Thus, we propose to estimate the cluster number for SSC via identifying the cluster centers based on the eigenvectors.

Recently, an effective cluster center identification method is proposed [24] in which a centrality coefficient metric is defined to measure the centrality of each point. And the cluster centers are identified as the points with extremely large centrality coefficients. Inspired by this, we first design a new metric, called centrality coefficient gap, to assess the ability of distinguishing cluster centers from other points under a given cluster number. Then, we estimate the number of clusters and the cluster centers by searching for the maximum centrality coefficient gap (See details in Section III-B). After achieving the cluster centers, we group the rest points into respective clusters which contain their nearest neighbors with higher local density [25], [24]. This leads to a new clustering algorithm for SSC. It not only sets the cluster number adaptively, but also eliminates the final randomly initialized k-means stage in traditional SSC, which often requires many iterations and depends highly on its initialization. We theoretically verify the correctness of the proposed cluster number estimation method on noise-free data. The experimental results on both synthetic and real-world data sets demonstrate the effectiveness of the proposed method on noisy data.

The rest of the paper is organized as follows. Section II introduces the preliminaries. Section III proposes the eigenvector analysis based cluster number estimation method and a new clustering algorithm for SSC. Section IV conducts experiments on both synthetic and real-world data sets. Finally, Section V concludes our work.

TABLE I: Common symbols used in the paper

Symbol	Description
$\boldsymbol{c}$	the sparse representation
$\mathcal{G}$	affinity graph
L	the normalized Laplacian graph matrix
λ	eigenvalue
v	eigenvector
ρ	the local density
δ	the minimum density-based distance
γ	the centrality coefficient
Δ	centrality coefficient gap
ϵ^2	noise level

II. Preliminaries

In this section, we give some background techniques related to our method. For clarity, Table I lists the symbols used throughout this paper.

A. Review of Sparse Subspace Clustering

Consider there is a collection of N data points $\boldsymbol{X} = \{\boldsymbol{x}_i \in \mathcal{R}^d\}_{i=1}^N$ lying in a union of l linear/affine subspaces $\{\mathcal{S}_i\}_{i=1}^l$ of intrinsic low dimensions $\{d_i\}_{i=1}^l$. If these data points are corrupted by noise, the following optimization program [12], [13] is employed to learn the sparse representation for each data point.

$$\min \beta \parallel \boldsymbol{c}_i \parallel_1 + \tfrac{1}{2} \parallel \boldsymbol{x}_i - \boldsymbol{X}\boldsymbol{c}_i \parallel_{\ell_2}^2, s.t. c_{i,i} = 0, \qquad (1)$$

where $\boldsymbol{c}_i = < c_{i,1}, c_{i,2}, \cdots, c_{i,N} >^T$, and $\beta > 0$ is a parameter to make a trade-off between sparsity and reconstruction error.

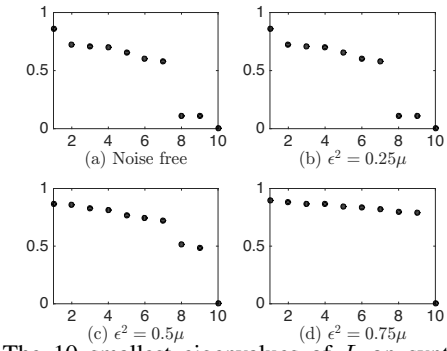


Fig. 1: The 10 smallest eigenvalues of L on synthetic data w.r.t. different noise levels

By solving the optimization program in Eq. 1, SSC gets a sparse representation $\boldsymbol{c}_i \in \mathbb{R}^N$ for each $\boldsymbol{x}_i$. Then, a matrix $\boldsymbol{C} \triangleq [\boldsymbol{c}_1;\ \boldsymbol{c}_2;\ \cdots;\ \boldsymbol{c}_N] \in \mathbb{R}^{N\times N}$ and a similarity matrix $W \triangleq |\boldsymbol{C}|+|\boldsymbol{C}|^T$ are constructed. Afterwards, SSC builds a weighted graph $\mathcal{G} \triangleq (\mathcal{V}, \mathcal{E}, W)$, where $\mathcal{V}$ denotes the nodes in $\mathcal{G}$ which correspond to N points of $\boldsymbol{X}$, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the edge set with weights given by W. Finally, it explores the clustering structure of $\boldsymbol{X}$ by applying spectral clustering on $\mathcal{G}$.

Under certain range of β, the solution of Eq. 1 can recover an approximate sparse representation with a bounded error [12], [13]. However, such approximation introduces a perturbation in the normalized Laplacian graph matrix L of $\mathcal{G}$, and thus the eigenvalues and eigenvectors of L vary according to matrix perturbation theory [26]. In practice, we cannot overlook such influence. It increases the challenge for cluster number estimation in SSC especially when the noise level is high.

Cluster Number Estimation for SSC: for noise-free data, there are ideally l components in $\mathcal{G}$ and no connections among them [17]. The multiplicity of the zero eigenvalue of L is equal to the number of clusters [17], [18], [27]. If the data are corrupted by noise, the eigenvalues of L will deviate from 0 [17], [27]. In this situation, one method frequently used for cluster number estimation in SSC is to search for the maximum drop in the magnitude of the eigenvalues [15], [16]. This approach, however, lacks a theoretical justification and is sensitive to the noise [18].

There have been a few eigenvector analysis based cluster number estimation methods for spectral clustering[18]. However, its performance largely relies on the similarity measure used to construct the affinity graph [17], [18], [23]. SSC measures the similarity among points based on the sparse coefficients over subspaces. And the connectivity within each subspace, which is built using such sparse coefficients, might be false when subspace dimension is greater than 3 [11]. This is quite different from the similarity used in the methods which compute the similarity among points in the original feature space. And the connectivity within each cluster is always true. Thus, it is debatable whether these methods works well for SSC.

B. Metrics for Cluster Center Identification

A proposition about the property of cluster centers is shown as follows [25]:

Proposition 1: Comparing to the other points, cluster centers have a higher local density and a relative larger distance from data points with higher local densities.

Based on this proposition, several methods have been proposed to identify the cluster centers recently [24], [25]. This section introduces the metrics used to identify the cluster centers in [24]: i) M-density, ii) Minimum density-based distance, and iii) Centrality coefficient.

Definition 1: For each point $\boldsymbol{x}_i \in \boldsymbol{X}$, let M be a given number of nearest neighbors of $\boldsymbol{x}_i$. M-density ρ of $\boldsymbol{x}_i$ is defined as follows.

$$\rho_i = M / \sum\nolimits_{j=1}^{M} d_{i,j}, \tag{2}$$

where $\{d_{i,j} | 1 \leq j \leq M\}$ denotes the set of distances between the data point $\boldsymbol{x}_i$ and its M nearest neighbors. If $\sum_{j=1}^{M} d_{i,j} = 0$, we set $\rho_i = M/EPS$, where EPS is the minimum distance that a floating point arithmetic program of a machine. Based on M-density, the minimum density-based distance δ of $\boldsymbol{x}_i$ is defined as follow.

Definition 2: Minimum density-based distance δ.

$$\delta_i = \min_{j \neq i \wedge \rho_i \leq \rho_j} (d_{i,j}) \tag{3}$$

The clustering centers usually have local maximum density, and so can be recognized as the points with anomalously large δ_i [25].

As the cluster centers have both higher M-density and minimum density-based distance, the centrality coefficient of $\boldsymbol{x}_i$ is defined as the product of these two metrics [24].

Definition 3: Centrality Coefficient γ.

$$\gamma_i = \rho_i \times \delta_i. \tag{4}$$

It has been demonstrated that γ follows a long-tailed distribution and the cluster centers lie at the end of the tail [24]. If we represent ρ and δ of each point in a two dimensional plot, clustering centers are usually far away from other points, and the clustering centers can be viewed as the outliers in the-two dimensional plot [24], [25]. Inspired by this observation, clustering centers are identified as the points corresponding to the outliers of γ followed a long-tailed distribution [24]. That is, the difference between the centrality coefficients of the cluster centers and those of the other points is large.

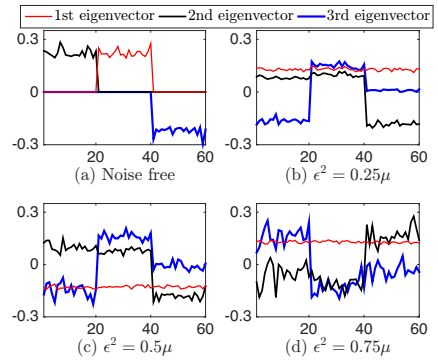
III. A Robust Self-Tuning Clustering Algorithm

In this section, we first give a motivating example in Section III-A, and design a new eigenvector analysis based method to estimate the cluster number in Section III-B. Then, based on the estimation method, we propose a new clustering algorithm for SSC in Section III-C.

A. A Motivating Example

A synthetic dataset is employed to demonstrate i) the robustness of eigenvectors on noise and ii) the motivation of our proposed cluster number estimation method.

Let $D \in \mathbb{R}^{1000 \times 60}$ be a noise-free data consisting of sixty 1000-dimensional points and each column has a unit Euclidean norm. There are three clusters lying in a union of linear 3D subspaces. The three clusters successively correspond to the first, middle and last 20 points, respectively. Let $\mu = \frac{1}{60 \times 1000} \sum_{i=1}^{1000} \sum_{j=1}^{60} |D_{i,j}|$ where $D_{i,j}$ is the jth value of ith data point, and μ is thus the average of all the absolute values in D. Then, we corrupt D by white noise $\mathcal{N}(0, \epsilon^2)$ in different levels. Specially, we generate a noise matrix $E \in \mathbb{R}^{60 \times 1000}$ where each element follows $\mathcal{N}(0, \epsilon^2)$, and set ϵ^2 as 0.25μ, 0.5μ, and 0.75μ, respectively. Finally, we get three corrupted data sets by $D + E$.



(a) Noise free (b) $\epsilon^2 = 0.25\mu$ (c) $\epsilon^2 = 0.5\mu$ (d) $\epsilon^2 = 0.75\mu$

Fig. 2: The eigenvectors wr.t. the smallest three eigenvalues on synthetic data

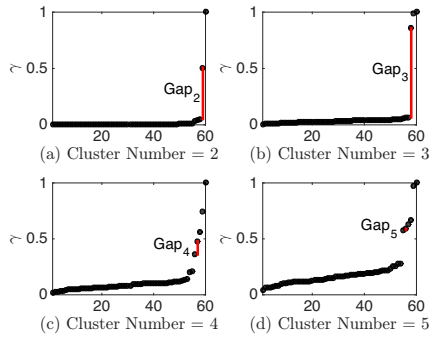
Fig. 1 shows the 10 smallest eigenvalues w.r.t. the noise-free data and three corrupted data. Fig. 2 shows the eigenvectors corresponding to the smallest three eigenvalues. From these figures, we can observe that:

(1) For noise-free data (See Fig. 1 (a)), the count of zero values is just the number of clusters. Meanwhile, the eigenvectors corresponding to the three zero eigenvalues have a regular stair-step shape (See Fig. 2 (a)), from which it is intuitive to distinguish three clusters via these eigenvectors.

(2) For noisy data, all eigenvalues except for the smallest one deviate from 0, and the perturbations appear in the values of eigenvectors. As the noise increases, a larger deviation of eigenvalue from 0 occurs and the perturbation in all eigenvectors becomes stronger. In this case, it is difficult to detect the correct cluster number via searching for either the count of zero eigenvalues or the maximum eigenvalue gap (see Fig. 1 (d) for example).

By comparison, when we resort to eigenvectors, although they are fluctuating, they still carry enough information to distinguish different clusters. For example, in Fig. 2 (d), the differences among the eigenvectors are still clear on three clusters even the fluctuation is strong. In particular, for the 1st cluster (first 20 points), the values in the 2nd eigenvector are always under those of the 1st and 3rd eigenvectors. The same phenomenons can be found over the 2nd and 3rd clusters. This kind of difference can be easily captured by k-means. And we can still get a perfect segmentations as long as all these three eigenvectors are picked up for clustering. Note that he same conclusion can be found on the synthetic data with more clusters. This demonstrates the eigenvectors are more robust to the noise comparing to eigenvalues.

Motivation: Taking data corrupted by noise $\epsilon^2 = 0.75\mu$ as



(a) Cluster Number = 2 (b) Cluster Number = 3 (c) Cluster Number = 4 (d) Cluster Number = 5

Fig. 3: The centrality coefficients γ w.r.t. different number of clusters on synthetic data

an example, we first compute the centrality coefficients of 60 points when the number of clusters/eigenvectors used for clustering is set as 2, 3, 4 and 5, respectively. Then, we normalize these coefficients with the infinite norm to make the centrality coefficient gaps comparable under different cluster numbers. We sort these centrality coefficients and draw them in Fig. 3.

If the cluster number is set as k, the points with top k centrality coefficients will be identified as the k cluster centers according to [24]. And the gap between the kth (corresponding to the point from cluster centers) and $(k-1)$th (corresponding to point from other non-center points) largest centrality coefficients can be used to evaluate the ability to distinguish cluster centers from the other points. We mark such gap by red line in Fig. 3.

From Fig. 3, we can observe the maximum gap is achieved when the number of clusters is set as the actual cluster number 3 (See Fig. 3 (b)). Inspired by this observation, we formulate the problem of cluster number estimation as an optimization program, where the objective is to search for the maximum centrality coefficient gap. See details in next section.

Remark 1: A very important and necessary step prior is to normalize the centrality coefficients by the infinite norm. This helps to better deal with different settings of cluster numbers and allows the centrality coefficient gaps computed under different settings to be comparable.

B. Cluster Number Estimation Method

Let $\lambda_1 \geq \lambda_2 \cdots \lambda_N$ be the eigenvalues of L in a descending order, and v_i $(1 \leq i \leq N)$ be the eigenvector corresponding to λ_i. Suppose k $(k > 1)$ is a given number of clusters for SSC, we calculate the centrality coefficients for all N points based on the last k eigenvectors, i.e., $\{v_{N-j+1}\}_{j=1}^{k}$, where the nearest neighbors of a point are identified based on Euclidean distance computed over these eigenvectors, and get $\boldsymbol{\gamma}^{(k)} = \gamma_1^{(k)}, \gamma_2^{(k)}, \cdots, \gamma_N^{(k)}$. Then, we normalize these coefficients as $\boldsymbol{\gamma}^{(k)} \leftarrow \boldsymbol{\gamma}^{(k)} / \parallel \boldsymbol{\gamma}^{(k)} \parallel_\infty$ according to Remark 1.

Let $\gamma_{(1)}^{(k)} \leq \gamma_{(2)}^{(k)} \leq \cdots \leq \gamma_{(N)}^{(k)}$ be the sorted normalized centrality coefficients of N points, where the subscript (i) represents the ith order $(1 \leq i \leq N)$. We define a new metric k-centrality coefficient gap $(k > 1)$ as follows.

Definition 4: k-Centrality Coefficient Gap.

$$\Delta_k = \gamma_{(k)}^{(k)} - \gamma_{(k-1)}^{(k)}. \tag{5}$$

Let K $(2 \leq K < N)$ be the maximal cluster number to be tested, and define the search space of the candidate cluster numbers as $[2, K]$. In practical application, as we do not know the actual cluster number in advance, we can set K as a large value to make sure the optimal solution falls in the search space. Then, we can estimate the cluster number via

$$k_{opt} = \arg\max_k (\Delta_k). \tag{6}$$

According Eq. 6, we compute the centrality coefficient gap for all candidate cluster numbers in range $[2, K]$, and pick up the number corresponding to the maximum centrality coefficient gap as the solution. Afterwards, we can identify the points with top k_{opt} centrality coefficients as the cluster centers. We introduce a theorem which demonstrates the correctness of Eq. 6 on noise-free data.

Theorem 1: Let D be a noise-free data lying in a union of l linear subspaces. The solution of optimization program in Eq. 6 for SSC on D is $k_{opt} = l$.

Proof: The proof consists of four steps:

Step 1: Eigenvector collection. As D is a noise-free data and lie in a union of l linear subspaces, there are ideally l connected components in the affinity graph $\mathcal{G}$ according to [8], [9]. Therefore, the normalized graph Laplacian matrix $L \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ is strictly block diagonal, and has blocks $L^{(i)}$ $(i = 1, 2, \cdots, l)$. The eigenvalues and eigenvectors of L are the union of the respective eigenvalues and eigenvectors of its blocks that are padded appropriately with zeros [23]. Then, the l eigenvectors corresponding to the l smallest igenvalues of L have ideally the following form: $\boldsymbol{V} = \begin{bmatrix} \boldsymbol{v}^{(1)} & \vec{\mathbf{0}} & \vec{\mathbf{0}} \\ \vec{\mathbf{0}} & \cdots & \vec{\mathbf{0}} \\ \vec{\mathbf{0}} & \vec{\mathbf{0}} & \boldsymbol{v}^{(l)} \end{bmatrix}$ where $\boldsymbol{v}^{(i)}$ is the eigenvector corresponding to the smallest eigenvalue of the sub-matrix $L^{(i)}$. SSC normalizes each row in $\boldsymbol{V}$ as a unit vector before clustering [8] and gets $\boldsymbol{V} = \begin{bmatrix} \mathbf{1}^{(1)} & \vec{\mathbf{0}} & \vec{\mathbf{0}} \\ \vec{\mathbf{0}} & \cdots & \vec{\mathbf{0}} \\ \vec{\mathbf{0}} & \vec{\mathbf{0}} & \mathbf{1}^{(l)} \end{bmatrix}$. Let $n_{\min}$ be the number of points in the smallest cluster. To compute the local density of each point over $\boldsymbol{V}$, the parameter of the number of nearest neighbors is selected as $M \in (1, n_{\min}]$ according to the setting in [24].

Step 2: Computation of Δ_l, i.e., the centrality coefficient gap with respect to optimal number of clusters. We first compute the local density and minimum density-based distance for each point over $\boldsymbol{V}$.

i) $\forall \boldsymbol{x}_i$, as $M \leq n_{\min}$, all of its M nearest neighbors come from the same cluster. The distances between $\boldsymbol{x}_i$ and its M neighbors are all zeros as each point has the same representation in $\boldsymbol{V}$. Therefore, all the points have same local density ρ according to Eq. 2.

ii) To compute the minimum density-based distance for $\boldsymbol{x}_i$, we first sort local densities of N points, i.e., $\rho_{(1)} \geq \rho_{(2)} \geq \cdots \geq \rho_{(N)}$. The corresponding points are denoted as $\{\boldsymbol{x}_{(i)}\}_{i=1}^{N}$, where $\boldsymbol{x}_{(i)}$ is the point with the (i)th density $\rho_{(i)}$. [1] The maximum distance among N point is $\sqrt{2}$. According

[1] As all N local densities have the same value, the orders of these N points can be in any order

to Definition 2, the minimum density-based distance of $\boldsymbol{x}_{(1)}$ is $\delta_{(1)} = \sqrt{2}$.

$\forall \boldsymbol{x}_{(i)}$ $(i > 1)$, if there are some points in $\{\boldsymbol{x}_{(j)}\}_{j=1}^{i-1}$ from the same cluster as $\boldsymbol{x}_{(i)}$, $\delta_{(i)} = 0$; otherwise, $\delta_{(i)} = \sqrt{2}$ according to Definition 2. After scanning all points, we get l points with $\delta = \sqrt{2}$, which are from l different clusters, and the rest $N-l$ points with $\delta = 0$.

iii) Considering all points have same local density, according to Definition 3, the l points with $\delta = \sqrt{2}$ have the same non-zero centrality coefficients. For the other $N - l$ points with $\delta = 0$, their centrality coefficient is 0. Therefore, $\Delta_l = 1$ according to Def. 4.

Step 3: Computation of Δ_k, $k \neq l$. This step consists of two situations:

i) $k < l$. We miss the last $(l - k)$ eigenvectors in $\boldsymbol{V}$. As $M \leq n^{(1)}$, all the points have the same M-density ρ.

Then, we can sort N densities as $\rho_{(1)} \geq \rho_{(2)} \geq \cdots \geq \rho_{(N)}$, and the point corresponding $\rho_{(i)}$ is denoted as $\boldsymbol{x}_{(i)}$. According to Definition 2, the minimum density-based distance of $\boldsymbol{x}_{(1)}$ is $\delta_{(1)} = \sqrt{2}$. It is noted that all points have the same local density. $\boldsymbol{x}_1$ can be from the following two situations: a) If $\boldsymbol{x}_{(1)}$ is from the first k clusters, according to Definition 2, by scanning all the data points in $\{\boldsymbol{x}_{(i)}\}_{i=1}^{N}$ in order, we can get that there are k points with $\delta = \sqrt{2}$, one point with $\delta = 1$ and the rest points with $\delta = 0$. Considering all points have the same local density, based on the Definitions 3 and 4, Δ_k = 1 - $1/\sqrt{2}$. b) If $\boldsymbol{x}_{(1)}$ is from the last $(l - k)$ clusters, according to Definition 2, there are k points with $\delta = 1$, one point with $\delta = \sqrt{2}$ and the rest points with $\delta = 0$. In this case, considering all N points have the same M-density, based on the Definitions 3 and 4, $\Delta_k = 0$. Therefore, $\Delta_k < \Delta_l$ when $k < l$.

ii) $k > l$. Superfluous $(k - l)$ eigenvectors are also selected for clustering. As the similarity matrix is strictly block diagonal, information contained in these $(k - l)$ eigenvectors is redundant to that in l optimal eigenvectors. Such redundant information results in intra-cluster distances are different and even larger than the inter-cluster distance. In this case, it is impossible that all the points have the same density ρ and the the minimum density-based distance will be quite complex according to Definitions 1 and 2. As we have normalized the centrality coefficients by the infinite norm, the gap of any pair of neighbors will be less than 1. Therefore, $\Delta_k < \Delta_l$ holds when $k > l$.

Step 4: In summary, no matter $k < l$ or $k > l$, it is always true that $\Delta_k < \Delta_l$. Therefore, the solution of Eq. 6 is l. ■

For data corrupted by noise, it is generally difficult to provide the theoretical analysis of the eigenvector distribution [27]. However, as long as the eigenvectors contain enough information to distinguish different clusters, Eq. 6 remains effective to estimate the correct cluster number for SSC. This can be illustrated by the example data occurred by noise in Section III-A. Figure 4 shows the centrality coefficient gaps when we set $K = 10$ over four example data sets. The estimated cluster number is marked by a red circle. From this figure, we can observe that Eq. 6 gives the correct estimation for all data sets.

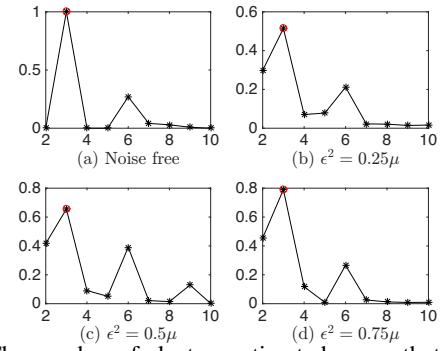

(a) Noise free
(b) $\epsilon^2 = 0.25\mu$
(c) $\epsilon^2 = 0.5\mu$
(d) $\epsilon^2 = 0.75\mu$

Fig. 4: The number of clusters estimated on synthetic data by Eq. 6 w.r.t. different noise levels

Time complexity: Let N be the number of data points and K be the maximum cluster number to be tested. According to the analysis in [24], for each centrality coefficient gap calculation, the time complexity to get the sorted centrality coefficients of N points for each given setting of number of clusters is $O(N^2)$. Thus, the computation cost of all K gaps is $O(N^2 \cdot K)$. We can find the maximum gap in $O(K)$, and thus estimate the number of clusters in $O(N^2 \cdot K) + O(K) = O(N^2 \cdot K)$.

To save the computation time, the search over the cluster number can be performed incrementally. Specifically, we start from the last two eigenvectors (i.e., corresponding to the two smallest eigenvalues), then, at each step of the search (up to the maximal cluster number K), we append a single eigenvector to update the already computed distances.

C. Robust Self-tuning SSC

Algorithm 1: Robust Self-tuning SSC

Inputs : X: $\{\boldsymbol{x}_i\}_{i=1}^{N}$ corrupted by noise;
K: the maximal cluster number to be tested, which defines the search space as [2, K];
Output: CLU;//A set of clusters

foreach $\boldsymbol{x}_i \in \boldsymbol{X}$ **do**
  $\boldsymbol{c}_i \leftarrow$ solution of optimization program in Eq.1;
$\boldsymbol{C} \leftarrow [\boldsymbol{c}_1; \boldsymbol{c}_2; \cdots; \boldsymbol{c}_N]$, $W \leftarrow |\boldsymbol{C}| + |\boldsymbol{C}|^T$;
Construct the affinity graph $\mathcal{G}$ over W;
$L \leftarrow$ normalized graph Laplacian matrix of $\mathcal{G}$;
$[S, V] \leftarrow SVD(L)$//S: eigenvalues, V: eigenvectors
for $k \leftarrow 2, 3, \cdots, K$ **do**
  $\boldsymbol{\gamma}^{(k)} \leftarrow$ compute centrality coefficients by Def. 3;
  $\Delta_k \leftarrow$ compute k-centrallity coefficient gap on $\boldsymbol{\gamma}^{(k)}$ by Eq. 5;
$k_{opt} = \arg\max_k(\Delta_k)$;
Identify the points w.r.t. top k_{opt} values in $\boldsymbol{\gamma}^{(k_{opt})}$ as cluster centers $\boldsymbol{O} = \{\boldsymbol{o}_1, \cdots, \boldsymbol{o}_{k_{opt}}\}$, and label $\boldsymbol{o}_i$ as i;
foreach $\boldsymbol{x}_i \in \boldsymbol{X} - \boldsymbol{O}$ **do**
  Mark $\boldsymbol{x}_i$ the label of its nearest neighbour with a higher density;
$CLU \leftarrow \{Clu_i\}_{i=1}^{k_{opt}}$, where Clu_i consists of points with label i;

Based on the cluster number estimation method in Section III-B, a new robust self-tuning SSC algorithm is proposed in Algorithm 1. In lines 1–6, we construct the affinity graph $\mathcal{G}$ based on the sparse representations of all points by solving the optimization program in Eq. 1. Then, we compute the eigenvalues S and eigenvectors V of the normalized Laplacian graph matrix L of $\mathcal{G}$ (line 7). Afterwards, we compute the centrality coefficient gap for each candidate cluster number in search space $[2, K]$ (lines 8–11), and estimate the cluster number as k_{opt} in line 12. Finally, we pick up the points with top k_{opt} centrality coefficients as the cluster centers (line 13), and then partition the rest points into the respective clusters containing their nearest neighbors with higher local density according to the clustering scheme in [24], [25] (lines 14–16). Note that this step eliminates the final iterative k-means process in traditional SSC, which often has to run for many iterations and depends highly on its initialization.

Complexity analysis: We omit the runtime for computing the sparse representations and eigenvalue decomposition of the normalized graph Laplacian matrix L which is a common step in any SSC algorithms. The analysis of time complexity of Algorithm 1 is shown as follows. In lines 8 – 12, estimating the cluster number is $O(N^2 \cdot K)$ according to the analysis in Section III-B. Line 13 infers k_{opt} cluster centers in $O(1)$ time complexity since we only need to pick up points with the top k_{opt} centrality coefficients. Finally, clustering the rest data points in lines 14 – 16 costs $O(N)$ [24]. Thus, the time complexity of the proposed SSC algorithm is $O(N^2 \cdot K) + O(N) = O(N^2 \cdot K)$.

IV. Experiments

Competitors i) The most commonly used eigenvalue analysis based method for SSC[16], [9], it estimates the number of clusters via searching for the maximum eigenvalue gap. We named this algorithm as SSC_{EG} (Eigenvalue Gap based SSC). ii) Eigenvector rotation based method ZP [18], which is the most popular cluster number estimation method in spectral clustering, and detects the number of clusters for spectral clustering by minimizing the cost function via rotating the eigenvectors. We denote this algorithm as SSC_{ZP}.

Our proposed algorithm estimates the cluster number based on Centrality Coefficient Gap, and is denoted as SSC_{CCG}.

The proposed algorithm and all the competitors are written in Matlab. All experiments are run on the same machine with an Intel Core Quad i5 with 2.6 GHz and 8GB RAM.

Parameter setting The parameters for approximate sparse representation calculation via Eq. 1 in all the three SSC algorithms, and the local density calculation in our proposed SSC_{CCG} are set according to their original papers. In paticular, we set the number of nearest neighbors M for local density calculation in SSC_{CCG} as $\sqrt{N}$ [24], where N represents the number of data points. Since all the competitors use k-means to explore the cluster structure over the selected eigenvectors, in each experiment, k-means is run 20 times with random starting points, and the maximum number of iterations is set as 200 [8].

TABLE II: Parameter Setting for Synthetic Data

Parameter (notation)	Settings	Default
Noise level (ϵ^2)	0.25μ, 0.5μ, 0.75μ,	0.5μ
Subspace dimension (d_s)	3, 5, 10	5

* μ is the average of all absolute values in the data.

Evaluation metrics We report the estimated cluster number, Normalized Mutual Information (NMI) [28], and runtime to compare different algorithms.

Note that the sparse representation calculation and eigenvalue decomposition on the normalized Laplacian graph matrix are common for all the three algorithms. Thus, we compared the runtime on cluster number estimation. We did not report the runtime of SSC_{EG} since the time on searching for the maximum eigenvalue gap can always be ignored comparing to the other two eigenvector analysis based methods.

A. Synthetic Data

1) Quality: To assess the performance of different methods on estimating cluster number for SSC, we systematically generated a set of synthetic data sets with different number of clusters. Meanwhile, there are two factors that lead to the difficulty for recovering sparse representations and clustering of SSC over corrupted data [16]: i) noise level ϵ^2 and ii) subspace dimension d_s. Generally, either a high noise level or a large subspace dimension increases the difficulty of SSC [16]. Therefore, we analyze the influence of ϵ^2 and d_s on different algorithms by varying one while fixing the other as default value. The setting of these two parameters is shown in Table II.

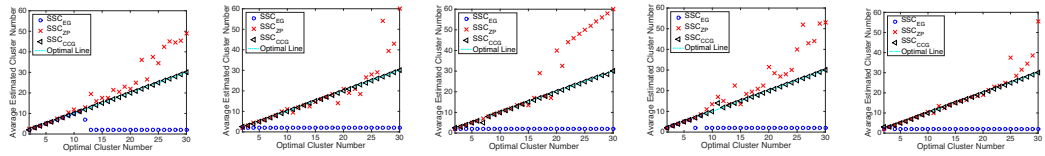
Specially, we generated 29 data sets (2000-dimensional) in which the number of clusters varies from 2 to 30 with step 1, and each cluster includes 30 points. To make a fair comparison on corrupted data, we added noise to each synthetic data under 10 different random seeds, and reported the average performance of each algorithm.

For each dataset, where the actual cluster number is l, we set the maximum cluster number to be tested $K = 2\times l$ to make sure that the optimal solution falls in the search space.

Impact of Noise: We fixed subspace dimension d_s as default value and set the noise level as 0.25μ, 0.5μ, and 0.75μ, respectively. We showed the comparison results of three SSC algorithms in terms of estimated cluster number in Figs. 5(a), 5(b), and 5(c), and NMI in Figs. 6(a), 6(b), and 6(c). From these figures, we can observe that:

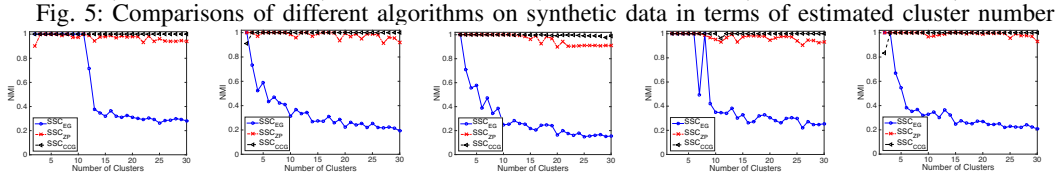
(1) The eigenvalue gap analysis based algorithm SSC_{EG} is sensitive to the noise. Its performance drops significantly with the increase of noise level ϵ^2. SSC_{EG} fails to detect the correct cluster number and results in poor NMI when the noise level is high. This is because of that, the greater noise introduces more connections among the points from different clusters, and results in the maximum eigenvalue gap occurs at the 1st and 2nd smallest eigenvalues and so the estimation of SSC_{EG} is always 2.

(2) Both of eigenvector analysis based algorithms SSC_{ZP} and SSC_{CCG} are robust to the noise comparing to SSC_{EG}. However, SSC_{ZP} usually tends to a larger estimation especially when the actual cluster number is great. This is because of that SSC_{ZP}



(a) $\epsilon^2 = 0.25\mu$, $d_s = 5$ (b) $\epsilon^2 = 0.5\mu$, $d_s = 5$ (c) $\epsilon^2 = 0.75\mu$, $d_s = 5$ (d) $\epsilon^2 = 0.5\mu$, $d_s = 3$ (e) $\epsilon^2 = 0.5\mu$, $d_s = 10$

Fig. 5: Comparisons of different algorithms on synthetic data in terms of estimated cluster number



(a) $\epsilon^2 = 0.25\mu$, $d_s = 5$ (b) $\epsilon^2 = 0.5\mu$, $d_s = 5$ (c) $\epsilon^2 = 0.75\mu$, $d_s = 5$ (d) $\epsilon^2 = 0.5\mu$, $d_s = 3$ (e) $\epsilon^2 = 0.5\mu$, $d_s = 10$

Fig. 6: Comparisons of different algorithms on synthetic data in terms of NMI

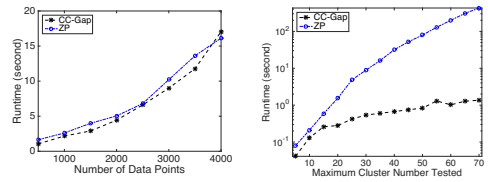
would estimate multiple cluster numbers yielding the same minimal cost. In this case, the largest of those is selected [18]. Meanwhile, to guarantee the optimal solution can be found, we set the upper bound of the search space as two times of the actual cluster number, the larger the actual cluster number is, the more candidates appearing in the search space. This increases the probability of multiple cluster numbers with same cost. Therefore, SSC_{ZP} favors the solution with large value when the actual cluster number is large. In comparison, the optimization program in our algorithm SSC_{CCG} still works well on the data when the actual cluster number is large.

Impact of Subspace Dimension: We set the subspace dimension d_s as 3, 5, and 10 respectively while fixing the noise level $\epsilon^2 = 0.5\mu$. We showed the comparison results of three SSC algorithms in terms of estimated cluster number (in Figs. 5(d), 5(b), and 5(d)) and NMI (in Figs. 6(d), 6(b), and 6(e)).

From these figures, we can observe that i) a larger subspace increases the difficulty of cluster number estimation for SSC_{EG}. When $d_s = 3$ (See Fig. 5(d)), SSC_{EG} can still get correct estimation if the actual cluster number is small (< 6) even the data are corrupted by a moderate-intensity noise. However, when $d_s = 5$ (See Fig. 5(b)) or 10 (See Fig. 5(b)), SSC_{EG} fails. ii) SSC_{ZP} still tends to picking up large estimation especially when the actual cluster number is large. iii) Our proposed algorithm SSC_{CCG} gets the best approximation to the optimal solution. Therefore, the eigenvector analysis based estimation method is robust to the subspace dimension and more practical since the real-world data usually lies in subspace dimension $d_s > 4$ [9].

In summary, our proposed algorithm SSC_{SSC} is robust to the noise and the dimension of subspace. The cluster number estimation method is more effective especially when the actual cluster number is large comparing to the competitors.

2) Scalability: We compared the runtime of our proposed Centrality Coefficient Gap (CC-Gap) based cluster number estimation method to ZP [18]. The time complexity of these methods is related to the sizes of data and the search space (i.e., the maximum cluster number to be tested). We first generated a 2000-dimension data with five clusters lying in a union of 5D subspaces where each cluster contains 100 points. Afterwards, i) to assess the runtime against the data size, we varied the data size from 500 to 4000 by a step size 500,



(a) Runtime v.s. # of Points (b) Runtime v.s. Search Space

Fig. 7: Scalability Analysis

and set the maximum cluster number to be tested as 20. ii) To assess the runtime against the size of the search space, we varied the number of cluster number to be tested from 5 to 70 with step 5. We fed each cluster number estimation method with the same affinity matrix. The runtime does not include time for the affinity matrix computation and eigenvalue decomposition.

The results are given in Fig. 7. From this figure, we can observe that i) as the data size increases, both methods take more time for cluster number estimation. And the variation trend of our proposed method CC-Gap is similar to that of ZP. ii) As the size of search space increases, CC-Gap is significantly faster than ZP especially for large setting (about two orders of magnitude). This means that our method CC-Gap is more practical than ZP since in practice, we do not know the actual cluster number in advance. To guarantee the optimal solution is included in the search space, we should set a large search space.

B. Real World Data

TABLE III: Summary of Real World Data

ID	Data Name	#Instance	#Feature	#Cluster	Area
1	GLIOMA*	50	4434	4	Biological
2	lung*	203	3312	5	Biological
3	Orlraws10P*	100	10304	10	Image
4	Pixraw10P*	100	10000	10	Image
5	Carcinom*	174	9182	11	Biological
6	YaleB◇	2414	1024	38	Image

* : http://featureselection.asu.edu/datasets.php.
◇ : http://www.cad.zju.edu.cn/home/dengcai/Data/FaceData.html.

In this section, we conduct experiments on six well-known real-world data sets collected from different areas (e.g., image and biological). Table III shows the summary information of the these data sets. For each data set with C clusters, we set the parameter, the maximum cluster number to be tested, as $2 \times C$.

TABLE IV: Comparison Results on Six Real World Data

ID	Cluster Number				NMI			Time (sec)	
	Opt	EG	ZP	CCG	EG	ZP	CCG	ZP	CCG
1	4	2	2	4	0.10	0.10	**0.18**	0.08	**0.06**
2	5	2	2	4	0.45	0.45	**0.66**	0.26	**0.07**
3	10	2	14	10	0.36	0.83	**0.93**	0.82	**0.05**
4	10	2	12	9	0.42	0.92	**0.94**	1.26	**0.05**
5	11	2	2	8	0.43	0.43	**0.77**	2.21	**0.07**
6	38	2	76	34	0.08	0.59	**0.69**	2282.17	**29.48**

* Opt: optimal cluster number; EG: SSC_{EG}; ZP: SSC_{ZP}; and CCG: SSC_{CCG}.

Comparisons on real data sets Table IV shows the comparison results of different algorithms on the real world data sets. From this table, we can get that i) for most of the data sets, $SSC_{CCG} > SSC_{ZP} > SSC_{EG}$ in terms of NMI. ii) SSC_{CCG} gets the estimation which is closer to the optimal cluster number comparing to the other methods. iii) SSC_{CCG} is more efficient especially when the size of data is large, such as for YaleB, SSC_{CCG} is faster than SSC_{ZP} around two orders.

In summary, comparing to the well-known competitors, the proposed algorithm SSC_{CCG} is more effective especially when the actual cluster number is large.

V. Conclusions and Future Work

Selecting correct cluster number to tune the clustering process of SSC is an art requiring skill and patience. Automating SSC is the main motivation for this study. In this paper, we have proposed a cluster number estimation method for SSC on corrupted data. The method is proposed based on these ideas: i) The eigenvectors are more robust to noise comparing to eigenvalues. ii) Estimating the optimal cluster number is equivalent to identifying a good set of cluster centers, and the optimal cluster number results in it is more intuitive to distinguish cluster centers from other points. Thus, we first designed a new metric, called centrality coefficient gap, to measure the discrimination between cluster centers and other points computed on eigenvectors. Then, we automatically selected the cluster number by maximizing such gap and got the cluster centers. Finally, we grouped the remaining points into clusters containing their nearest neighbors with higher density. We theoretically verified the correctness of our method on noise-free data. Extensive experiments and evaluations on both of synthetic and real-world data show our method works well for SSC on data corrupted by noise.

Future work would go to 1) speed up the compuation of cluster number selection by exploiting the sparsity or introducing indexing structures, and 2) extend the cluster number estimation method to other spectral clustering based methods, e.g., Low-Rank Subspace Clustering (LRSC) [29] and Low-Rank Sparse Subspace Clustering (LRSSC) [30].

Acknowledgments

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